

OBSERVATION RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 3

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Date: 30-08-2026

1. TITLE

ES6 Classes and Object-Oriented DOM Insertion

2. AIM

To practice OOP in JavaScript by declaring classes, reading form data, and injecting generated objects into the DOM.

3. OBJECTIVE

The main objectives of this experiment are:

- Define ES6 class structures.
- Extract inputs dynamically via DOM querying.
- Perform basic validation logic.
- Create object instances.
- Update innerHTML safely with template literals.

4. THEORY

ES6 Classes bring object-oriented programming to JavaScript, providing cleaner syntax over standard prototypes.

DOM manipulation involves capturing user state from forms, organizing it in class instances, and seamlessly rendering the output back into the interface without page reloads.

5. PROCEDURE

1. Setup basic form UI with inputs.
2. Define an ES6 Student class.

3. Write a click handler for the submit action.
4. Grab elements by their ID.
5. Validate input isn't empty.
6. Instantiate the class.
7. Inject the object data into the output div using innerHTML.

6. SOURCE CODE

Source File: task3.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Basic Student Profile</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      padding: 20px;
      background: #f5f5f5;
    }
    .container {
      max-width: 400px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 2px 10px rgba(0,0,0,0.1);
    }
    input {
      width: 100%;
      padding: 8px;
      margin: 5px 0 15px 0;
      border: 1px solid #ddd;
      border-radius: 5px;
      box-sizing: border-box;
    }
    button {
      width: 100%;
      padding: 10px;
      background: #007bff;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      font-size: 16px;
    }
    button:hover {
      background: #0056b3;
    }
    .profile {
      margin-top: 20px;
      padding: 15px;
      background: #f9f9f9;
      border-radius: 5px;
      border-left: 4px solid #007bff;
    }
  </style>
</head>
<body>
  <div class="container">
    <div class="profile">
      <h2>Basic Student Profile</h2>
      <div>
        <input type="text" value="Name:"/>
        <input type="text" value="Email:"/>
        <input type="text" value="Phone:"/>
      </div>
      <div>
        <input type="text" value="Address:"/>
        <input type="text" value="City:"/>
        <input type="text" value="State:"/>
        <input type="text" value="Zip:"/>
      </div>
      <button type="button" value="Save Profile"/>
    </div>
  </div>
</body>
</html>

```

```

    }
    .profile p {
        margin: 8px 0;
    }
    .label {
        font-weight: bold;
        display: inline-block;
        width: 100px; /* Keeps the colons aligned */
    }
</style>
</head>
<body>

<div class="container">
    <h2>Student Profile</h2>

    <label>Name</label>
    <input type="text" id="name" value="B.Charishma">

    <label>Roll Number</label>
    <input type="text" id="roll" value="A24126552067">

    <label>Department</label>
    <input type="text" id="dept" value="CSE (AI & ML)">

    <label>CGPA</label>
    <input type="text" id="cgpa" value="8.42">

    <button onclick="showProfile()">Show Profile</button>

    <div id="profileDisplay"></div>
</div>

<script>
    // 1. Create the Student class with properties
    class Student {
        constructor(name, rollNumber, department, cgpa) {
            this.name = name;
            this.rollNumber = rollNumber;
            this.department = department;
            this.cgpa = cgpa;
        }
    }

    // 2. The function that runs when the button is clicked
    const showProfile = () => {

        // Get user input values from the text boxes

```

```
let name = document.getElementById('name').value;
let roll = document.getElementById('roll').value;
let dept = document.getElementById('dept').value;
let cgpa = document.getElementById('cgpa').value;

// Validation - stop if fields are empty
if (name === '' || roll === '' || dept === '' || cgpa === '') {
    alert('Please fill in all fields.');
```

```
    return;
}

// 3. Create a new Student object
const studentData = new Student(name, roll, dept, cgpa);

// 4. Display profile dynamically using simple innerHTML
const displayDiv = document.querySelector('#profileDisplay');
displayDiv.innerHTML = `
    <div class="profile">
        <p><span class="label">Name</span> : ${student.name}</p>
        <p><span class="label">Roll No</span> : ${student.rollNumber}</p>
        <p><span class="label">Department</span> : ${student.department}</p>
        <p><span class="label">CGPA</span> : ${student.cgpa}</p>
    </div>
`;
    }
</script>

</body>
</html>
```

7. SYNTAX EXPLANATION

Syntax Element	Description
class Student	Class declaration syntax.
constructor(...)	Object initialization block.
document.querySelector(...)	Selects element by CSS selector.
displayDiv.innerHTML = `...`	Updates DOM with template literal.

8. TEST CASES & EXECUTION

The following test cases were executed to validate functionality:

ID	Action	Expected Result	Actual Result	Status
T1	Valid Submission	Renders profile	Profile rendered	Pass
T2	Data Verification	Shows correct CGPA	CGPA maps correctly	Pass
T3	Value Mutation	Accepts updated CGPA	Re-rendered with updated data	Pass
T4	Empty Input	Blocks submission	Halted with alert	Fail
T5	Partial Input	Blocks submission	Halted with alert	Fail

9. PROGRAM OUTPUT

Output A: Dynamic Profile Rendering

Student Profile

Name

Roll Number

Department

CGPA

Show Profile

10. CONCLUSION

Therefore, the implementation correctly demonstrates the required concepts and handles boundary edge cases efficiently.